



CSE 200 • B2 • Group 2

VARIATIONAL AUTOENCODER

Learning to generate, not just compress

Presented by **Member 1** • **Member 2** • **Member 3**

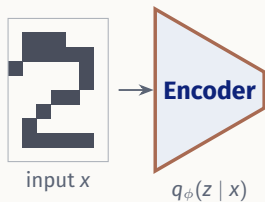
Department of CSE, BUET



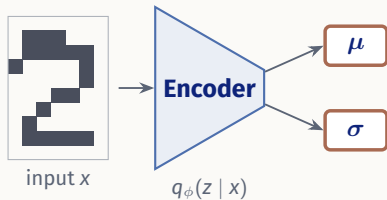
latent space $z \sim \mathcal{N}(\mu, \sigma^2)$



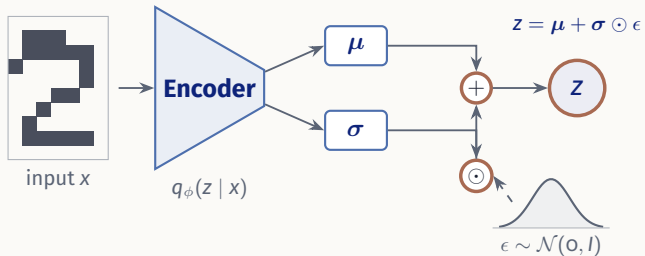
The VAE Pipeline: $x \rightarrow z \rightarrow \hat{x}$



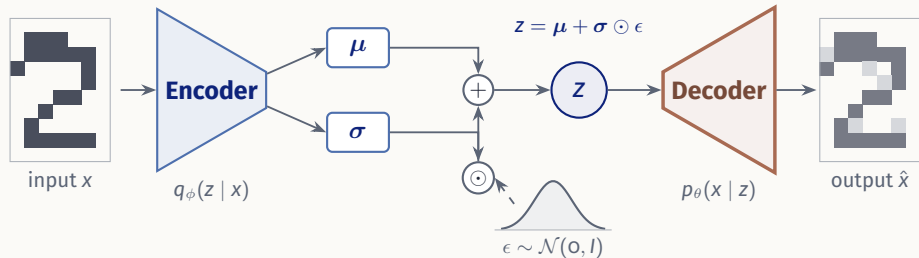
The VAE Pipeline: $x \rightarrow z \rightarrow \hat{x}$



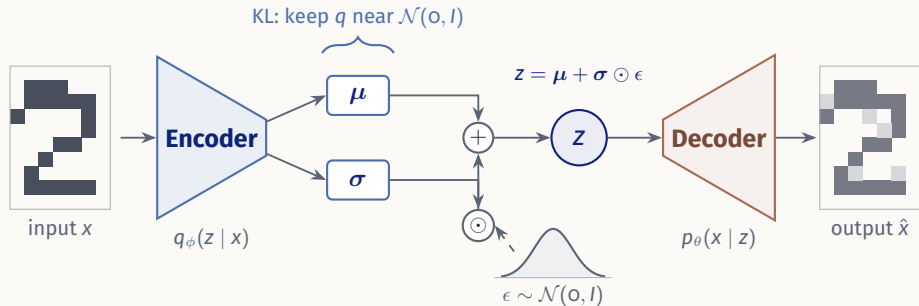
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$$\mathcal{L}(\theta, \phi; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x | z)]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z | x) \| p(z))}_{\text{regularisation}}$$